Customer Shopping Behavior Analysis

Problem Overview

This project analyzes customer shopping behavior using Python, PostgreSQL, SQL, and Power BI to identify sales patterns, customer preferences, and business insights. The objective was to transform raw shopping data into actionable insights that can support better business decisions.

Exploratory Data Analysis using Python

We began with data preparation and cleaning in python

- **Data Loading:** Imported dataset using Python.
- **Initial Exploration:** Used **df.info()** to check structure and **df.describe()** for summery statistics.
- **Missing data handling:** Checked for null values and imputed missing values in the Review **Rating column** using median rating for each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned dataframe into the database for SQL analysis.

Data Analysis using SQL

Which gender contribute the highest revenue - Male customers contribute the highest sale.

	gender	total_revenue
0	Male	157890.0
1	Female	75191.0

Which group spends the most - Middle age contribute the highest revenue among all age groups

	age_group	total_revenue
0	Middle Age	67711.0
1	Adult	64927.0
2	Young Adult	53140.0
3	Senior	43164.0
4	Teen	4139.0

Which location generate the highest sale?

	location	total_revenue
0	Montana	5784.0
1	Illinois	5617.0
2	California	5605.0
3	Idaho	5587.0
4	Nevada	5514.0
5	Alabama	5261.0

Which product category most popular among different age group - Clothing is the most popular category among all.

	age_group	category	total_purchases
0	Adult	Clothing	489
1	Middle Age	Clothing	492
2	Senior	Clothing	320
3	Teen	Clothing	33
4	Young Adult	Clothing	403

Top 5 products by rating ?

	item_purchased	avg_rating
0	Gloves	3.86
1	Sandals	3.84
2	Boots	3.82
3	Hat	3.80
4	Skirt	3.78

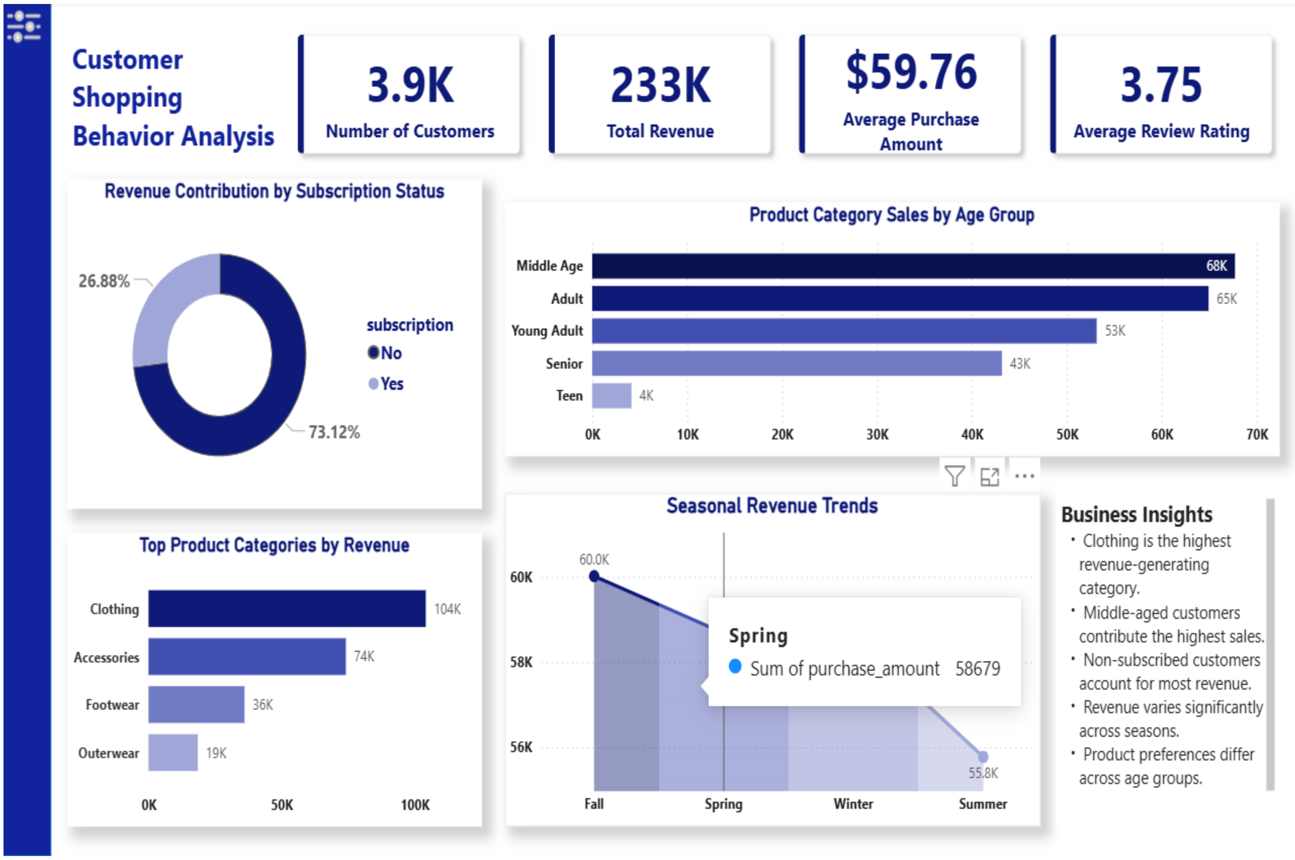
Do subscribed customers make more purchases - Non-Subscribed customers demonstrate higher purchasing activity and spending compared to subscribed customers, indicating stronger customer engagement.

	subscription_status	total_purchases	avg_purchase
0	No	2847	59.87
1	Yes	1053	59.49

Do discounts increase purchase amount?

	discount_applied	total_orders	avg_purchase_amount
0	No	2223	60.13
1	Yes	1677	59.28

Dashboard in Power Bi



Business Recommendation

1. **Improve Subscription Conversion -** Non-subscribed customers generate the majority of revenue, indicating an opportunity to convert high-value customers into subscribers through personalized offers and loyalty benefits.
2. **Focus on High-Revenue Categories -** The business should prioritize inventory, promotions, and marketing efforts toward top-performing product categories to maximize revenue growth.
3. **Target Age-Based Marketing -** Different age groups prefer different product categories, so personalized marketing campaigns should be designed for each customer segment.
4. **Improve Customer Satisfaction** - Highly rated products should receive increased visibility and promotion, while low-rated products should be reviewed for quality improvements.